

S1 Text. Parameter Mapping, Scoring Rubric, and Illustrative Regime Scores

Parameter-to-Observable Mapping Tables

0.1 Mapping Table

Legibility Parameters

Enforcement Parameters

Exit Parameters

Institutional Parameters

Dimension Rubrics and Illustrative Regime Scores

0.2 Dimension Rubrics

Legibility (L)

- 0: Compliance primarily assessed through internal dispositions; external acts are weakly diagnostic of moral standing. - 1: Some external markers exist but institutional evaluation emphasizes internal criteria. - 2: Mixed. Institutional evaluation uses both external and internal criteria without clear dominance. - 3: External markers carry primary institutional weight; internal states are acknowledged but secondary in practice. - 4: Moral classification terminates at visible compliance. External performance is treated as sufficient evidence of moral standing.

Substitutability (S)

- 0: External acts explicitly insufficient for moral standing; institutional discourse insists on interior transformation. - 1: External acts necessary but consistently described as preliminary to interior work. - 2: External acts necessary; interior transformation endorsed but

Table 1: Legibility parameters: model-to-observable mapping.

Model Parameter	Definition	Observable Proxies	Scoring Anchors
σ (substitution weight)	Weight of external compliance relative to internal cultivation in perceived moral score: $m_i = (1-\sigma)x_i + \sigma s_i$	Proportion of canonical prescriptions targeting observable behavior vs. internal states; institutional emphasis on orthopraxy vs. orthodoxy of intention	$\sigma \approx 0.2$: moral evaluation primarily references internal dispositions (e.g., Theravāda emphasis on mental cultivation). $\sigma \approx 0.5$: mixed; both external and internal criteria carry institutional weight. $\sigma \approx 0.8$: moral classification terminates at visible compliance (e.g., dress code adherence as primary social marker of piety).
v_{obs} (visibility of external compliance)	Probability that an observer accurately perceives an agent's external compliance signal	Density of public ritual requirements; frequency of communal worship; visibility of identity markers in daily life; institutional surveillance mechanisms	$v_{\text{obs}} \approx 0.3$: compliance largely private; few public markers. $v_{\text{obs}} \approx 0.9$: compliance continuously visible through dress, attendance, dietary practice, or communal monitoring.
a_{obs} (observability of internal states)	Probability that an observer perceives an agent's internal cultivation	Institutional mechanisms for assessing sincerity (confession, spiritual direction, examination of conscience, teacher-student relationship with behavioral criteria)	$a_{\text{obs}} \approx 0.05$: internal states are essentially opaque to observers (default for most large-scale systems). $a_{\text{obs}} \approx 0.3$: institutional mechanisms partially reveal internal states (e.g., monastic communities with long-term teacher observation).

not institutionally enforced. - 3: External acts widely treated as sufficient in institutional practice, even if doctrinal sources acknowledge interior requirements. - 4: External compliance fully substitutes for moral standing. Institutional mechanisms evaluate and reward

Table 2: Enforcement parameters: model-to-observable mapping.

Model Parameter	Definition	Observable Proxies	Scoring Anchors
π (enforcement reward)	Status, material, or institutional payoff to agents who punish noncompliance	Clerical promotion contingent on orthodoxy enforcement; social prestige for public correction of deviants; material compensation for informing; institutional access granted to enforcers	$\pi \approx 0$: no institutional reward for policing (enforcement is purely costly). $\pi \approx 0.15$: moderate status gains for boundary maintenance. $\pi \approx 0.3$: enforcement is a reliable career path with compounding institutional rewards.
λ (punishment severity)	Magnitude of sanction imposed on detected noncompliers	Codified penalties for infractions: fines, shunning, excommunication, corporal punishment, imprisonment, execution. Informal sanctions: gossip, ostracism, marriage exclusion	$\lambda \approx 0.05$: mild social disapproval. $\lambda \approx 0.15$: significant reputational and economic consequences (shunning, exclusion from communal resources). $\lambda \approx 0.3$: severe sanctions including physical punishment or legal penalties.
θ (heresy threshold)	Moral score below which an agent is classified as deviant and subject to sanctioning	Clarity and rigidity of orthodox/heterodox boundary; existence of formal heresy categories; specificity of infraction definitions	θ high (strict): narrow band of acceptable belief/practice; minor deviations trigger enforcement. θ low (lax): broad tolerance; only extreme deviation triggers response.

proxy performance without reference to internal state.

Enforcement Affordance (A)

- 0: Deviations difficult to define; violations interpretively ambiguous; no institutional sanction pathway. - 1: Some identifiable infractions but sanctions are informal and inconsistent. - 2: Defined infraction categories exist; sanctions available but unevenly applied. - 3: Clear binary violations with established institutional sanction processes. - 4: Comprehensive infrac-

Table 3: Exit parameters: model-to-observable mapping.

Model Parameter	Definition	Observable Proxies	Scoring Anchors
Exit cost (composite)	Social, economic, and identity penalties for leaving the system; modeled as a function of network degree, threat level, and base opportunity	Legal apostasy penalties; kinship network endogamy rate; economic dependence on community institutions; geographic isolation; identity foreclosure (single-identity vs. layered-identity environments)	Low: departure is socially tolerated; intermarriage and cross-membership common; economic life largely external (e.g., liberal Protestant denominations in pluralistic urban settings). Medium: significant reputational cost; family strain; partial economic dependence. High: apostasy carries legal sanctions, family severance, or economic ruin (e.g., ex-communication with total shunning; apostasy laws with criminal penalties).

tion taxonomy with mandatory sanctions; enforcement infrastructure dedicated to detection and punishment.

Centralization of Adjudication (C)

- 0: Highly pluralistic; multiple legitimate interpretive authorities; no adjudicative monopoly.
- 1: Decentralized but with informal hierarchy (e.g., respected scholars without binding authority).
- 2: Federated authority; regional or denominational adjudication with limited cross-jurisdictional reach.
- 3: Hierarchical authority with recognized appellate structure; interpretive plurality constrained.
- 4: Monopolistic adjudication; single institutional body defines orthodoxy and controls sanctions.

Exit Cost (E)

Table 4: Institutional parameters: model-to-observable mapping.

Model Parameter	Definition	Observable Proxies	Scoring Anchors
Delegation (monopoly_on)	Whether punishment authority is concentrated in endogenously selected agents or distributed across the population	Existence of specialized enforcement roles (inquisitors, morality police, rabbinical courts, sharia judges, party commissars, honor enforcers); proportion of sanctions issued by institutional vs. ad hoc agents	Off: any member can sanction; enforcement is peer-to-peer. On: recognized institutional roles control sanctioning; non-selected agents are blocked or discouraged from independent enforcement.
Enforcement capital (cap)	Compounding stock of institutional resources, legitimacy, and organizational capacity that increases future enforcement throughput	Tenure and promotion of enforcement personnel; institutional memory and precedent systems; organizational infrastructure for surveillance and adjudication; budgetary growth of enforcement apparatus	Low: enforcement is episodic and does not compound. High: enforcement apparatus grows, professionalizes, and acquires self-sustaining institutional momentum.
Literalism (L)	Agent-level trait weighting strict textual interpretation in enforcement decisions	Proportion of enforcement justified by direct textual citation vs. contextual or analogical reasoning; institutional emphasis on literal reading of canonical sources; existence of literalist-contextualist institutional divide	L low: enforcement is justified through interpretive reasoning, contextual adaptation, or institutional discretion. L high: enforcement is justified by direct appeal to canonical text, with minimal interpretive mediation.

- 0: Departure socially tolerated; intermarriage and cross-membership common; minimal sanctions for leaving. - 1: Mild social disapproval; some reputational cost; economic and family life largely unaffected. - 2: Moderate reputational and social penalties; family strain; partial economic dependence on community. - 3: Severe social consequences: shunning, family severance, marriage market exclusion; significant economic cost. - 4: Legal penalties for apostasy; total community severance; physical danger or criminal prosecution.

0.3 Illustrative Regime Scores

The following scores are provisional and serve to demonstrate the rubric’s application. They are subject to revision through the inter-rater protocol described above. Each entry identifies a historically bounded institutional configuration, not a tradition as a whole.

Table 5: Illustrative regime scores for historically bounded institutional configurations.

Regime	L	S	A	C	E	Predicted Model Regime
Taliban governance, Afghanistan (1996–2001)	4	4	4	3	4	Capture
Saudi CPVPV enforcement era (1979–2016)	4	3	4	3	3	Capture
Counter-Reformation Roman Inquisition (1542–1700)	3	3	4	4	3	Capture
Haredi community, contemporary Bnei Brak	3	3	3	2	3	Mixed/Capture boundary
CCP party discipline system (post-1949)	4	4	4	4	4	Capture
Stalinist self-criticism sessions (1930s)	4	4	4	4	4	Capture
Theravāda forest tradition, Thailand (20th c.)	1	0	1	0	1	Quiet
Quaker meeting, contemporary US	1	0	0	0	0	Quiet
Mainline Protestantism, contemporary US	2	2	1	1	0	Collapse (slow attrition)
Early Scientology, Sea Org (1967–1986)	4	4	4	4	4	Capture
Sufi tariqas, Ottoman era (decentralized)	2	1	1	0	1	Quiet/Mixed boundary
Iranian morality police era (post-1979)	4	3	4	4	3	Capture